

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Cloud and Orchestrate”

Project Tile – Intelligent Shopping Agent

Domain of Project – Artificial Intelligence & Agentic AI

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Problem Statement -

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Intelligent Shopping Agent

Shoppers face an overwhelming number of product choices, scattered reviews, and changing prices across different e-commerce platforms. Finding the right product that balances **price, quality, ratings, reviews, and personal preferences** can be time-consuming and confusing.

Traditional recommendation systems often fail to understand **real-time trends, personalized requirements, budget limitations, urgency, and sustainability preferences**.

Therefore, there is a need to develop an **AI-powered Intelligent Shopping Agent** that can collect information from multiple sources, understand user requirements, compare products intelligently, and provide **personalized, real-time shopping recommendations**.

- **The Challenge**

Shoppers today face an overload of product options, scattered reviews, and constantly changing prices across different e-commerce platforms. Finding the right product while balancing **price, quality, ratings, personal preferences, and availability** can be time-consuming and confusing.

Traditional recommendation systems may not effectively consider **real-time trends, personalized requirements, budget, urgency, or sustainability**. Users often need to visit multiple platforms and manually compare products, prices, and reviews before making a decision

- **The Objective**

The main objective is to build an **Intelligent AI Shopping Agent** that provides **personalized, intelligent, and real-time shopping assistance**.

The system should:

1. **Fuse Multisource Data**

Collect and combine product details, prices, ratings, reviews, and availability from multiple e-commerce sources into unified information.

2. **Enable Multimodal Search**

Allow users to search using **text, product images, or natural-language descriptions** and identify relevant products.

Proposed Solution -

Proposed Solution: AI-Powered Nutrition Agent

We propose to develop an **AI-powered Intelligent Shopping Agent** using **IBM Langflow, IBM watsonx Orchestrate, and IBM Models** to provide users with a complete and personalized shopping experience. The agent will gather product information from multiple e-commerce sources and combine **prices, specifications, ratings, reviews, availability, and offers** into unified insights. Users will be able to search for products through **text, voice, or images**, making the system flexible and easy to use. The AI will understand the user's **budget, preferences, purpose, required features, and urgency** and recommend the most suitable products accordingly. It will compare products based on **price, quality, customer feedback, and value for money**, while also providing alternative choices. The system can analyze review patterns, identify common customer opinions, provide price and deal insights, and notify users about potential price drops when reliable data is available. A **Smart Shopping Pulse dashboard** will display product comparisons, personalized recommendations, trending products, sustainability information, and shopping lists in a simple and interactive format. By combining **multisource data, multimodal search, AI reasoning, predictive insights, and agentic workflows**, the proposed solution aims to reduce shopping time, simplify product research, and help users make **smarter, faster, and more confident purchasing decisions**.

Technology Used

- **IBM Langflow** – For designing and connecting AI workflows.
- **IBM watsonx Orchestrate** – For managing and coordinating AI agents and tasks.
- **IBM Models** – For natural-language understanding, reasoning, and recommendations.
- **Python** – For backend logic, data processing, and AI integration.
- **HTML, CSS & JavaScript** – For developing the interactive web interface and dashboard.
- **APIs / Web Data Sources** – For collecting product prices, ratings, reviews, and availability.
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- **Database** – For storing product information and user preferences.
- **Multimodal AI** – For supporting text, voice, and image-based product searches.

Orchestrate component Used -

1. **AI Agent** – Understands user requirements and manages the shopping process.
2. **LLM / IBM Model** – Analyzes queries and generates intelligent responses.
3. **Tools & APIs** – Fetch product prices, ratings, reviews, and availability.
4. **Flows / Workflows** – Connect different shopping tasks into an automated process.
5. **Decision Logic** – Compares products and selects the most suitable options.
6. **Data Processing** – Combines information from multiple sources.
7. **Recommendation Component** – Generates personalized product recommendations.
8. **User Interaction** – Handles text, voice, and image-based shopping queries.
9. **Notification / Alert** – Provides price-drop and deal alerts when supported by available data.

Agent Instructions Screenshot-

IBM watsonx Orchestrate

EnglishContact SalesBuy Now

30 trial days left

Intelligent Shopping Agent

BuildEvaluateDeploy

Create version

DraftLive

Select a voice mode for the agent to enable voice.

Voice mode is not selected

Embed on your website

Test your agent easily. Copy and paste this snippet into your environment to preview before deploying. [Learn more](#)

```
<script>
window.wx0Configuration = {
  orchestrationID: "ba6546472113476bba29ccd205b750af_22288d89-b0f9-4c...",
  hostURL: "https://au-syd.watson-orchestrate.cloud.ibm.com",
  rootElementID: "root",
  deploymentPlatform: "ibmcloud",
  crn: "crn:v1:bluemix:public:watsonx-orchestrate:au-syd:a/ba65464721...",
  chatOptions: {
    agentId: "f391f365-2999-4273-9918-fa87a88808a7",
  }
};
setTimeout(function () {
  const script = document.createElement('script');
  script.src = `${window.wx0Configuration.hostURL}/wxochat/wxoLoader...`;
  script.addEventListener('load', function () {
    // ...
  });
});
```

Show more

Slack

Configure Slack to connect with the agent in each environment.

Draft Preview

Save as test

Running in debug mode

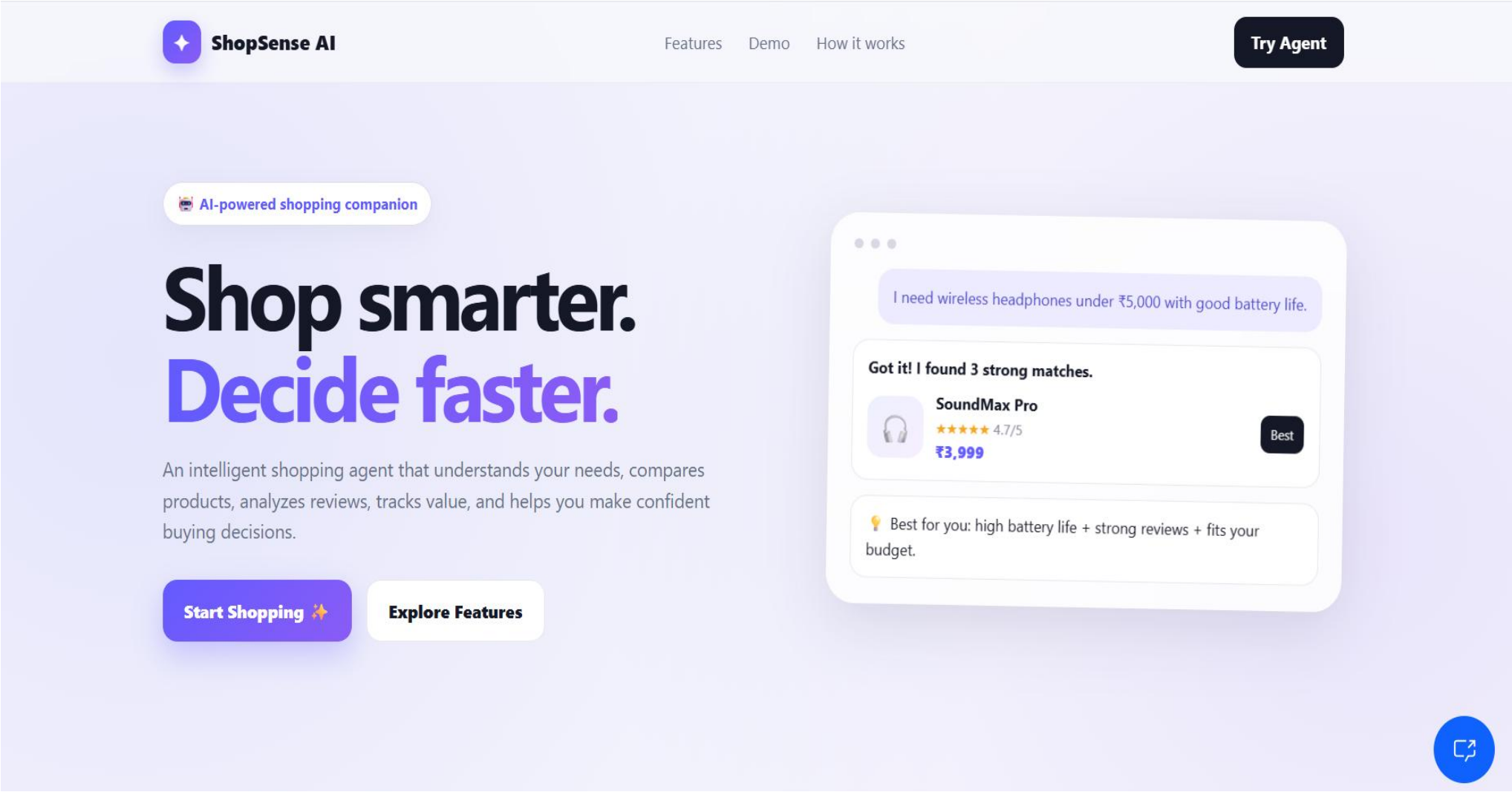
Intelligent Shopping Agent 4:36 PM

Sure! I can help you find the right décor items for your room. To give you the most useful suggestions, could you let me know a few details?

What I need to know	Why it helps
Budget range (e.g., total ₹5,000 – ₹20,000 or per-item limit)	Lets me prioritize value-for-money options.
Room type & size (living room, bedroom, study, square-feet)	Determines appropriate sizes and quantities.
Style / theme (modern, boho, minimalist, industrial, vintage, etc.)	Helps pick items that match your aesthetic.
Key items you're looking for (wall art, cushions, lighting, rugs, storage, plants, etc.)	Focuses the search on the products you need most.
Urgency (need it within a week, flexible)	Influences availability and shipping suggestions.
Sustainability preference (eco-friendly, recycled)	Allows me to highlight greener options.

dinning table under the 1000\$

Document Uploaded Screenshot (If applicable)-



Prompt with output Screenshot:

ShopSense AI

Features Demo How it works

AI-powered shopping companion

Shop smarter. Decide faster.

An intelligent shopping agent that understands your needs, compares products, analyzes reviews, tracks value, and helps you make confident buying decisions.

Start Shopping

Explore Features

Try Agent

Intelligent Shopping Agent

AI

Today

You 4:42 PM

Formal T shirts

Intelligent Shopping Agent 4:42 PM

Sure, I can help you find formal-style T-shirts that suit your needs. To give you the most relevant recommendations, could you let me know a few details?

What I need to know	Why it helps
Budget (per shirt or total)	Lets me filter by price and find the best value-for-money options.
Preferred material (e.g., cotton, polyester-blend, modal)	Determines comfort, drape, and suitability for formal settings.

Type something...

I need wireless headphones under ₹3,000.

Got it! I found 3 strong matches.

SoundMax Pro

★★★★★ 4.7/5

₹3,999

Best for you: high battery life + strong bass.

Other Applicable Screenshots:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>ShopSense AI | Intelligent Shopping Agent</title>
  <meta name="description" content="ShopSense AI is an intelligent shopping agent that compares products, reviews, prices and preferences." />

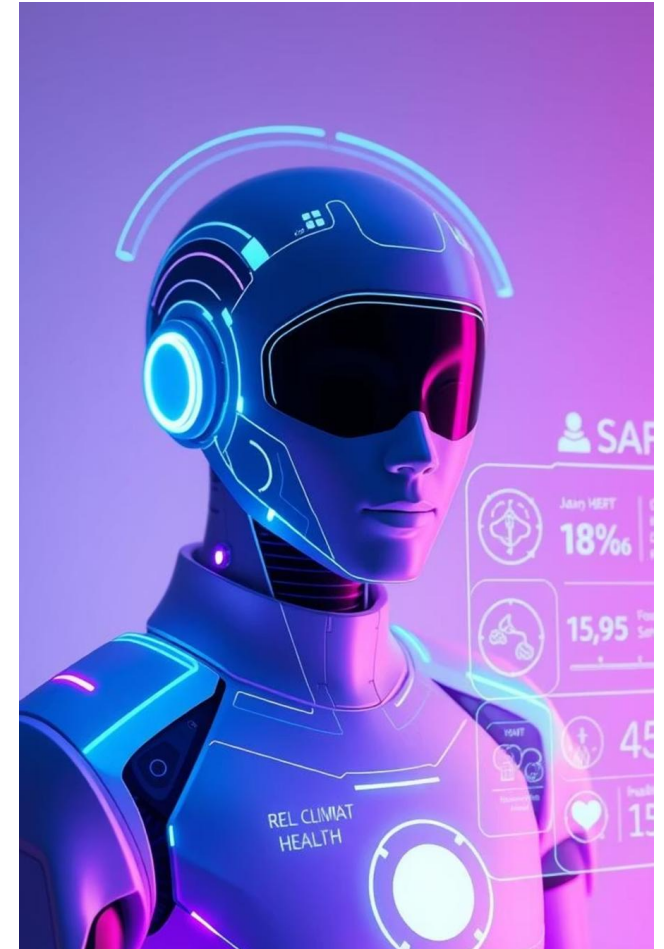
<style>
  * { box-sizing: border-box; margin: 0; padding: 0; }
  :root {
    --bg: #f7f8fc;
    --card: rgba(255,255,255,.86);
    --text: #151827;
    --muted: #687087;
    --primary: #635bff;
    --secondary: #8b5cf6;
    --border: #e7e8f0;
    --shadow: 0 18px 55px rgba(36, 31, 82, .12);
  }
  html { scroll-behavior: smooth; }
  body {
    font-family: Inter, ui-sans-serif, system-ui, -apple-system, BlinkMacSystemFont, "Segoe UI", sans-serif;
    background:
      radial-gradient(circle at 10% 10%, rgba(99,91,255,.12), transparent 28%),
      radial-gradient(circle at 90% 15%, rgba(139,92,246,.12), transparent 25%),
      var(--bg);
    color: var(--text);
    line-height: 1.6;
  }
  a { color: inherit; text-decoration: none; }
  button, input, select { font: inherit; }
  .container { width: min(1160px, 92%); margin: auto; }

  nav {
    position: sticky; top: 0; z-index: 20;
    backdrop-filter: blur(16px);
    background: rgba(247,248,252,.78);
    border-bottom: 1px solid rgba(231,232,240,.8);
  },
```

Role of Agentic AI in the solution

Role of Agentic AI in the Solution

Agentic AI acts as the **intelligent decision-making layer** of the shopping system. It understands the user's requirements, collects product information from multiple sources, compares prices, features, ratings, and reviews, and identifies the most suitable products. It can also analyze user preferences, suggest alternatives, provide deal and price insights, and support personalized recommendations. Using **IBM watsonx Orchestrate**, the agent coordinates different tools and workflows to complete shopping tasks automatically, while **IBM Models** provide reasoning and natural-language understanding. This makes the system more than a simple search engine—it works as a **smart shopping companion** that helps users save time and make informed purchasing decisions.



Novelty and Uniqueness

Novelty and Uniqueness

The proposed **Intelligent Shopping Agent** stands out by combining **Agentic AI, multisource data, multimodal search, predictive intelligence, and personalization** into a single intelligent shopping system.

Agentic AI Intelligence – Uses AI agents to understand user needs, compare products, and support purchase decisions.

Multisource Data Fusion – Combines prices, specifications, ratings, reviews, and availability from multiple sources.

Multimodal Search – Supports **text, voice, and image-based** product searches.

Hyper-Personalization – Tailors recommendations based on **budget, preferences, purpose, and urgency**.

Smart Price Intelligence – Provides deal insights and potential price-drop alerts using available price data.

Review Intelligence – Summarizes customer reviews and identifies important feedback patterns.

Predictive Shopping Support – Helps users decide whether to buy now or consider waiting when sufficient historical data is available.

Smart Shopping Pulse – Provides a unified dashboard for **product comparisons, trends, sustainability information, and personalized shopping lists**.

Scalable AI Architecture – Built using **IBM Langflow, IBM watsonx Orchestrate, and IBM Models** for AI workflows and agent coordination.

This makes the solution a **smart, personalized, and proactive shopping companion**, rather than just a traditional product search or recommendation tool.

Git Hub Link

Github URL – <https://github.com/coder-adiix>

❑ **Project Repository** – Intelligent Shopping Agent

- **Explore the source code, project implementation, documentation, and future development of our Intelligent Shopping Agent.**

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Future Scope

1. **Advanced AI Personalization and Predictive Shopping** – In the future, the agent can learn from users' shopping preferences, budgets, previous searches, and product interests to provide highly personalized recommendations. It can analyze historical price trends and market patterns to predict **discounts, price drops, seasonal offers, and the best time to purchase**, helping users save both time and money.
2. **Complete Autonomous Multi-Agent Shopping Ecosystem** – The system can be expanded with specialized AI agents for **product discovery, price comparison, review analysis, sustainability, and deal tracking**. These agents can work together through IBM watsonx Orchestrate to handle complex shopping tasks, while advanced **voice, image, and multilingual support** can make the platform accessible to a wider range of users.

Certificates Earned:

In recognition of the commitment to achieve professional excellence



Aditya kalambe

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



Issued on: Sep 02, 2026
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/fbde850d-bedb-462c-ba23-f129e795b799>



Certificates Earned:

IBM SkillsBuild	Completion Certificate
	This certificate is presented to Aditya kalambe for the completion of Lab: Troubleshoot Your Code Using IBM Bob (ALM-COURSE_4071307) According to the Adobe Learning Manager system of record
Completion date: 03 Sep 2026 (GMT)	Learning hours: 30 mins

Thank You!

Thank you for your time and interest.